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Homo Sapiens Chromosomal Location Ontology: A Framework for Genomic Data in Biomedical Knowledge Graphs

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The Homo sapiens Chromosomal Location Ontology (HSCLO) is designed to facilitate the integration of human genomic features into biomedical knowledge graphs from releases GRCh37 and GRCh38 at multiple resolutions. HSCLO comprises two distinct versions, HSCLO37 and HSCLO38, each tailored to its respective human genome release. This ontology supports the efficient integration and analysis of human genomic data across scales ranging from entire chromosomes to individual base pairs, thereby enhancing data retrieval and interoperability within large-scale biomedical datasets. Unlike existing ontologies that primarily focus on genomic feature identification or annotation, HSCLO is specifically engineered to optimize the interoperability and scalability of genomic data within biomedical knowledge graphs. The utility and performance of HSCLO are demonstrated through a case study involving the integration of high-resolution chromatin interaction data, which reveals significant improvements in query efficiency and data linkage. HSCLO represents a valuable resource for advancing research in disease genetics, personalized medicine, and other domains that require complex genomic data integration.

Introduction

Knowledge graphs (KGs) have become important tools in integrating and analyzing heterogeneous biomedical data, enabling the linkage of diverse datasets in a structured and semantically enriched manner. However, integrating genomic feature data into KGs presents significant challenges due to the need for precise alignment across varying chromosomal resolutions, from entire chromosomes to individual base pairs. Current methods for identifying overlapping genomic features can involve computationally intensive searches, particularly when applied to large-scale biomedical knowledge graphs containing millions of data nodes. This highlights the need for an ontological framework that could readily be integrated and utilized in knowledge graphs (in the form of subject/predicate/object triples) that standardizes the representation of chromosomal locations and reduces computational complexity during data integration.

In biomedical research, analyzing chromosomal locations of genomic features and their positional relationships is critical for understanding disease mechanisms, guiding therapeutic development, and personalizing medical treatments. An ontology that systematically catalogs and categorizes chromosomal positions is essential for supporting these research endeavors. For example, a researcher may wish to rapidly interrogate a knowledge graph for all genomic features and annotations that are proximal to a chromosomal variant. Such a framework would enable these queries by linking variants to associated genes, regulatory elements, or chromatin interaction data across different resolution levels, thereby expediting complex analyses and supporting hypothesis generation and experimental validation.

Existing ontologies have contributed significantly to the standardization of genomic data, particularly in annotating specific genomic features or chromosomal structures[1–4]. However, these ontologies are not optimized for integrating positional information at multiple chromosomal resolutions within KGs, with a standing gap in the ability to link genomic data across different resolution scales efficiently.

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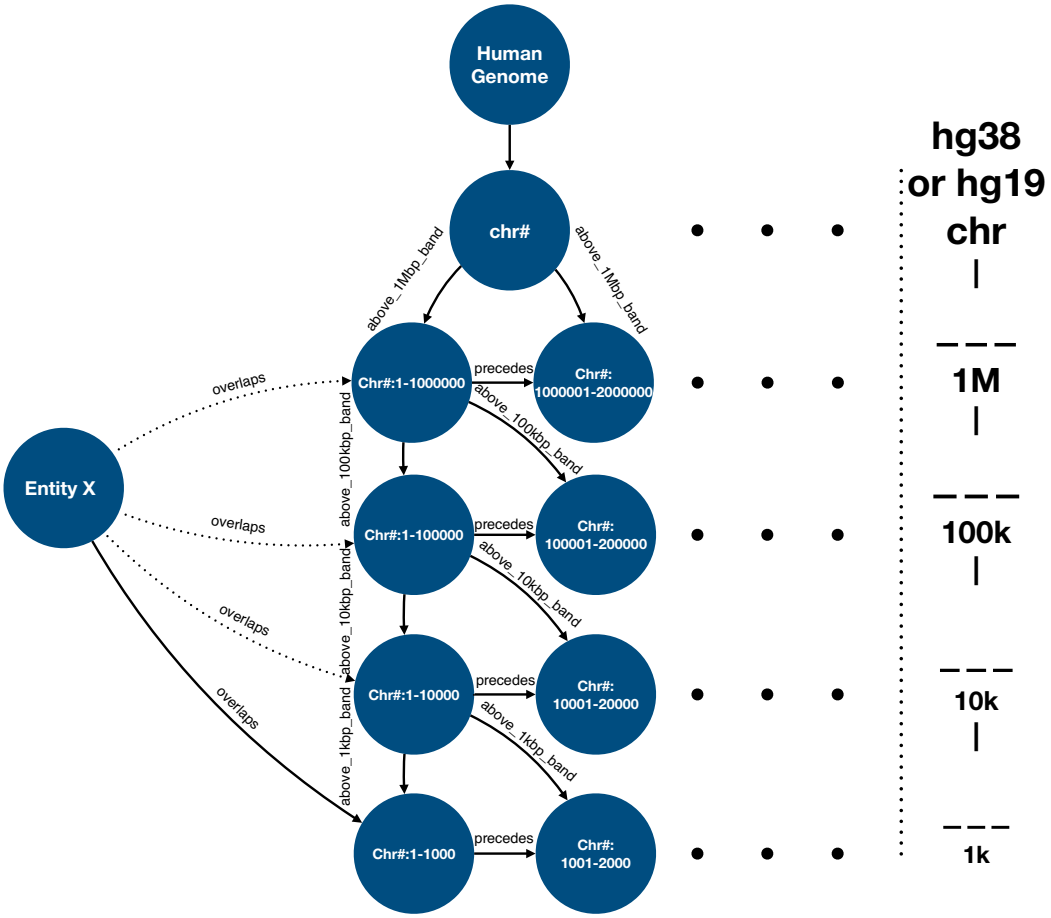

Fig. 1 Graphical Schema of HSCLO. Entity X could be any chromosomal feature, including chromosomal bands, genes, exons, introns, regulatory elements, QTLs, variants, accessible chromatin regions, viral integration sites, human endogenous retroviruses, transposons, tandem repeats, chromosomal contact regions, TADs, telomere, centromeres, and any other type.

To address this gap, we introduce the Homo sapiens Chromosomal Location Ontology (HSCLO), developed for the GRCh37[5] (HSCLO37) and GRCh38[6] (HSCLO38) genome assemblies (Fig. 1). HSCLO is specifically designed to facilitate the integration of positional information into biomedical knowledge graphs, supporting rapid querying and efficient data linkage across diverse experimental datasets. In this study, we detail the construction of HSCLO, present its application in a knowledge graph environment, and demonstrate its utility through a case study involving high-resolution chromatin interaction data integration.

The HSCLO is currently utilized in two knowledge graph projects, Petagraph[7] and the Common Fund Data Ecosystem's Data Distillery Project[8]. In the Data Distillery project, HSCLO has been indispensable in integrating Common Fund genomic datasets across different resolution levels, for example, in connecting 4DN chromosomal loop data[9] to individual genes or variants from sources such as GTEx[10] or Kids First[11].

Methods

Building HSCLO versions. To construct the HSCLO versions for GRCh37 and GRCh38, we downloaded each genomic coordinate file from the UCSC website (https://hgdownload.soe.ucsc.edu/goldenPath/hg19/bigZips/hg19.chrom.sizes and https://hgdownload.soe.ucsc.edu/goldenPath/hg38/bigZips/hg38.chrom.sizes). We parsed them into binned location nodes that were organized by chromosome size and scale using custom R scripts (https://github.com/TaylorResearchLab/HSCLO/blob/main/Codes/HSCLO38.R). The nodes were defined at five resolution levels (1 kbp, 10 kbp, 100 kbp, 1 Mb, chromosome) and connected through hierarchical relationships reflecting their chromosomal positioning. Each level in the ontology was linked to both its parent level and neighboring nodes using specific edge names (e.g., "`above_1Mbp_band`," "`precedes_10kbp_band`"). Nodes were named using a UCSC coordinate format-like structure[12] as for example HSCLO38:chrN:nnnn-nnnn. The preprocessing scripts are available on GitHub (https://github.com/TaylorResearchLab/HSCLO/blob/main/Codes/HSCLO38.R).

Use-case methodology. For our use case, we show how we establish the intersection between the 4DN dataset and Gencode to profile the overlap between 4DN loops and human genes in GRCh38. Once ingested in a knowledge graph in the Neo4j database management environment (Neo4j Desktop Community Edition 1.5.7), we used the Cypher query language to query ENSEMBL nodes in Gencode[12] and their corresponding HSCLO

Chromosome	1Mbp nodes	100kbp nodes	10kbp nodes	1kbp nodes	above_ 1Mbp_band	above_ 100kbp_band	above_ 10kbp_band	above_ 1kbp_band	precedes_ 1Mbp_band	precedes_ 100kbp_band	precedes_ 10kbp_band	preceedes_ 1kbp_band
1	249	2490	24896	248957	249	2490	24896	248957	248	2489	24895	248956
2	243	2422	24220	242194	243	2422	24220	242194	242	2421	24219	242193
3	199	1983	19830	198296	199	1983	19830	198296	198	1982	19829	198295
4	191	1903	19022	190215	191	1903	19022	190215	190	1902	19021	190214
5	182	1816	18154	181539	182	1816	18154	181539	181	1815	18153	181538
6	171	1709	17081	170806	171	1709	17081	170806	170	1708	17080	170805
7	160	1594	15935	159346	160	1594	15935	159346	159	1593	15934	159345
8	146	1452	14514	145139	146	1452	14514	145139	145	1451	14513	145138
9	139	1384	13840	138395	139	1384	13840	138395	138	1383	13839	138394
10	134	1338	13380	133798	134	1338	13380	133798	133	1337	13379	133797
11	136	1351	13509	135087	136	1351	13509	135087	135	1350	13508	135086
12	134	1333	13328	133276	134	1333	13328	133276	133	1332	13327	133275
13	115	1144	11437	114365	115	1144	11437	114365	114	1143	11436	114364
14	108	1071	10705	107044	108	1071	10705	107044	107	1070	10704	107043
15	102	1020	10200	101992	102	1020	10200	101992	101	1019	10199	101991
16	91	904	9034	90339	91	904	9034	90339	90	903	9033	90338
17	84	833	8326	83258	84	833	8326	83258	83	832	8325	83257
18	81	804	8038	80374	81	804	8038	80374	80	803	8037	80373
19	59	587	5862	58618	59	587	5862	58618	58	586	5861	58617
20	65	645	6445	64445	65	645	6445	64445	64	644	6444	64444
21	47	468	4671	46710	47	468	4671	46710	46	467	4670	46709
22	51	509	5082	50819	51	509	5082	50819	50	508	5081	50818
X	157	1561	15605	156041	157	1561	15605	156041	156	1560	15604	156040
Y	58	573	5723	57228	58	573	5723	57228	57	572	5722	57227
mtDNA	0	0	2	17	—	—	—	17	—	-	1	16

Table 1. Node and relationship statistics for HSCLO38, detailing the hierarchical organization of chromosomal locations in the GRCh38 genome assembly. The table includes counts for nodes at each genomic resolution level (1 Mbp, 100 kbp, 10 kbp, and 1 kbp) and their corresponding hierarchical ("above") and positional ("precedes") relationships for chromosomes 1–22, X, Y, and mtDNA.

nodes at 1kbp resolution using the query in Fig. 2. The query utilizes the mapping of ENSEMBL genes and transcript onto the HSCLO38 through the GENCODE_HSCLO mapping dataset:

```
//Query1: GENCODE Overlap with HSCLO38 at 1kbp
MATCH (o1:Code {SAB:'ENSEMBL'})<-[:CODE]-(c1:Concept)-[r1:us_5_prime]->
(c2:Concept)-[:CODE]->(o2:Code{SAB:'HSCLO'}),(c1:Concept)-
[r2:ds_3_prime]->(c3:Concept)-[:CODE]->(o3:Code {SAB:'HSCLO'}),
p=shortestPath((c2)-[:precedes_1kbp_band*]->(c3)) WHERE ((c2)<>(c3))
RETURN * LIMIT 1
```

Loop dotcall files of the following 4DN datasets were obtained from the 4D Nucleome project website data repository (https://data.4dnucleome.org). The dataset URLs and descriptions are provided in Table 3. The following query can identify 4DN loops and their anchor coordinates along HSCLO38 at 1kbp resolution.

```
//Query2: 4DN Overlap with HSCLO38 at 1kbp
MATCH (o1:Code {SAB:'4DNL'})<-[a:CODE]-(c1:Concept)-[r1]->
(c2:Concept)-[b:CODE]->(o2:Code {SAB:'HSCLO'}),(c1:Concept)-[r2]->
(c3:Concept)-[c:CODE]->(o3:Code {SAB:'HSCLO'}),p=shortestPath((c2)-
[:precedes_1kbp_band*]->(c3)) WHERE ((c2)<>(c3)) RETURN * LIMIT 1
```

Note that the query uses the Neo4j shortestPath procedure to extract HSCLO38 nodes connected using the "`precedes_1kbp_band`" relationship. Subsequently, genes that partially or entirely overlapped with human loops were identified through querying for shared links to HSCLO38:

```
//Query3: 4DN Overlap with GENCODE through HSCLO at 1 kbp resolution
MATCH (o1:Code {SAB:'4DNL'})<-[a:CODE]-(c1:Concept)-
[r1:loop_us_start]->(c2:Concept)-[b:CODE]->(o2:Code {SAB:'HSCLO'})
MATCH (c1:Concept)-[r2:loop_ds_end]->(c3:Concept)-[c:CODE]->(o3:Code
```

Chromosome	1Mbp nodes	100kbp nodes	10kbp nodes	1kbp nodes	above_ 1Mbp_band	above_ 100kbp_band	above_ 10kbp_band	above_ 1kbp_band	precedes_ 1Mbp_band	precedes_ 100kbp_band	precedes_ 10kbp_band	precedes_ 1kbp_band
1	250	2493	24926	249251	250	2493	24926	249251	249	2492	24925	249250
2	244	2432	24320	243200	244	2432	24320	243200	243	2431	24319	243199
3	199	1981	19803	198023	199	1981	19803	198023	198	1980	19802	198022
4	192	1912	19116	191155	192	1912	19116	191155	191	1911	19115	191154
5	181	1810	18092	180916	181	1810	18092	180916	180	1809	18091	180915
6	172	1712	17112	171116	172	1712	17112	171116	171	1711	17111	171115
7	160	1592	15914	159139	160	1592	15914	159139	159	1591	15913	159138
8	147	1464	14637	146365	147	1464	14637	146365	146	1463	14636	146364
9	142	1413	14122	141214	142	1413	14122	141214	141	1412	14121	141213
10	136	1356	13554	135535	136	1356	13554	135535	135	1355	13553	135534
11	136	1351	13501	135007	136	1351	13501	135007	135	1350	13500	135006
12	134	1339	13386	133852	134	1339	13386	133852	133	1338	13385	133851
13	116	1152	11517	115170	116	1152	11517	115170	115	1151	11516	115169
14	108	1074	10735	107350	108	1074	10735	107350	107	1073	10734	107349
15	103	1026	10254	102532	103	1026	10254	102532	102	1025	10253	102531
16	91	904	9036	90355	91	904	9036	90355	90	903	9035	90354
17	82	812	8120	81196	82	812	8120	81196	81	811	8119	81195
18	79	781	7808	78078	79	781	7808	78078	78	780	7807	78077
19	60	592	5913	59129	60	592	5913	59129	59	591	5912	59128
20	64	631	6303	63026	64	631	6303	63026	63	630	6302	63025
21	49	482	4813	48130	49	482	4813	48130	48	481	4812	48129
22	52	514	5131	51305	52	514	5131	51305	51	513	5130	51304
X	156	1553	15528	155271	156	1553	15528	155271	155	1552	15527	155270
Y	60	594	5938	59374	60	594	5938	59374	59	593	5937	59373
mtDNA	0	0	2	17	0	0	0	17	0	0	1	16

Table 2. Node and relationship statistics for HSCLO37, detailing the hierarchical organization of chromosomal locations in the GRCh37 genome assembly. The table includes counts for nodes at each genomic resolution level (1 Mbp, 100 kbp, 10 kbp, and 1 kbp) and their corresponding hierarchical ("above") and positional ("precedes") relationships for chromosomes 1–22, X, Y, and mtDNA.

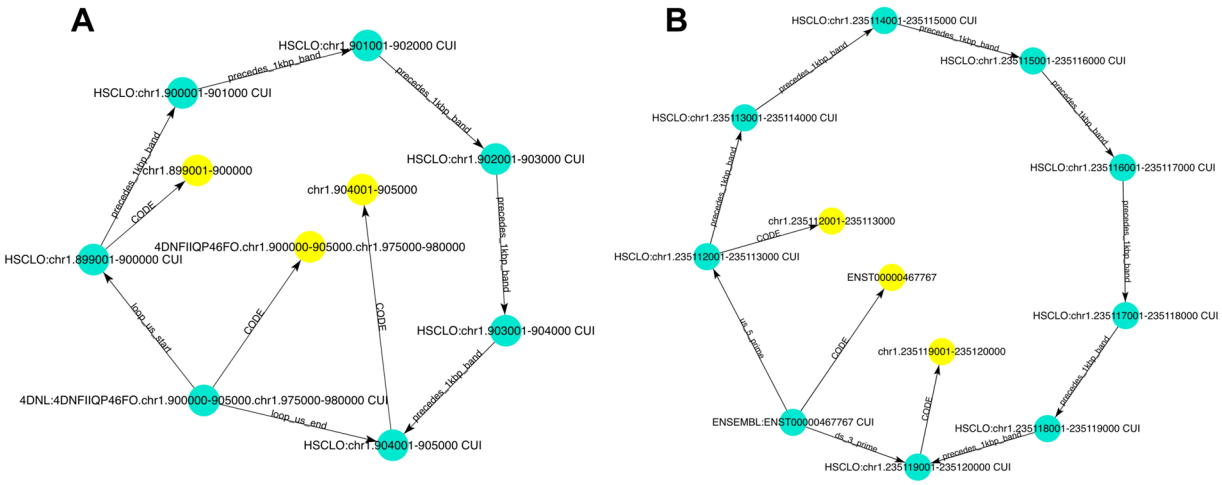

Fig. 2 Chromosomal features annotated with HSCLO. Example query results derived from Queries 1 and 2 outlined in the Methods section depict a 4DN loop upstream anchor (**A**) and a human gene transcript (**B**) as spanned along the HSCLO38.

```
{SAB:'HSCLO'}),p=shortestPath((c2)-[:precedes_1kbp_band*]->(c3))
WITH o1.CODE AS loop, nodes(p) AS P MATCH (o1:Code
{SAB:'ENSEMBL'})<-[:CODE]-(c1:Concept)-[r1]->(c2:Concept) WHERE (c2 in P)
 RETURN loop, o1.CODE
```

File Accession	Dataset Description	URL
4DNFIEY8UT8A	*in situ* Hi-C on HCT-116 (HCT-116-RAD21-mAC)	https://data.4dnucleome.org/files-processed/4DNFIEY8UT8A/
4DNFIOQLL854	Replicates of *in situ* Hi-C on untreated HCT116 WAPL-mAID2 cells	https://data.4dnucleome.org/files-processed/4DNFIOQLL854/
4DNFILP7BD5H	Replicates of *in situ* Hi-C on Auxin treated HCT116 WAPL-mAID2 cells	https://data.4dnucleome.org/files-processed/4DNFILP7BD5H/
4DNFILKW1NH2	Hi-C experiments on H1-hESC crosslinked with both FA and DSG, digested with DpnII.	https://data.4dnucleome.org/files-processed/4DNFILKW1NH2/
4DNFI3GNGT17	Hi-C experiments on H1-hESC crosslinked with both FA and DSG, digested with DpnII.	https://data.4dnucleome.org/files-processed/4DNFI3GNGT17/
4DNFIEEF14ST	Hi-C experiments on H1-hESC crosslinked with both FA and DSG, digested with DpnII.	https://data.4dnucleome.org/files-processed/4DNFIEEF14ST/
4DNFI4PMUZNU	Hi-C experiments on H1-hESC crosslinked with both FA and DSG, digested with DpnII.	https://data.4dnucleome.org/files-processed/4DNFI4PMUZNU/
4DNFI8NCHTIR	Hi-C on HFFc6 (Tier 1) with DpnII	https://data.4dnucleome.org/files-processed/4DNFI8NCHTIR/
4DNFIVO3FNLK	Hi-C on HFFc6 (Tier 1) with DpnII	https://data.4dnucleome.org/files-processed/4DNFIVO3FNLK/
4DNFIY71465K	Hi-C on HFFc6 (Tier 1) with DpnII	https://data.4dnucleome.org/files-processed/4DNFIY71465K/
4DNFILSK2QNY	Hi-C on HFFc6 (Tier 1) with DpnII	https://data.4dnucleome.org/files-processed/4DNFILSK2QNY/
4DNFIWEN3L7Y	Hi-C on HFFc6 (Tier 1) with DpnII	https://data.4dnucleome.org/files-processed/4DNFIWEN3L7Y/
4DNFIIQP46FO	MicroC Libraries from HFFc6 cells with 3 biological replicates, FA and DSG cross-linked, MNase digestion	https://data.4dnucleome.org/files-processed/4DNFIIQP46FO/
4DNFI212GCZW	Hi-C experiments on HFFc6 crosslinked with both FA and DSG, digested with DpnII.	https://data.4dnucleome.org/files-processed/4DNFI212GCZW/
4DNFI2K9STD5	Hi-C experiments on HFFc6 crosslinked with both FA and DSG, digested with DdeI.	https://data.4dnucleome.org/files-processed/4DNFI2K9STD5/
4DNFIQ48U7UJ	Hi-C experiments on HFFc6 crosslinked with both FA and DSG, digested with DdeI and DpnII.	https://data.4dnucleome.org/files-processed/4DNFIQ48U7UJ/
4DNFI3RMWQ85	MicroC Libraries from H1ESC cells from 2 biological replicates, FA and DSG cross-linked, MNase digestion	https://data.4dnucleome.org/files-processed/4DNFI3RMWQ85/
4DNFI3O1OTN4	*in situ* Hi-C on H1-hESC (Tier 1) with DpnII	https://data.4dnucleome.org/files-processed/4DNFI3O1OTN4/
4DNFI49A9Z1K	Micro-C experiments on H1 derived endoderm, differentiated using the 4DN protocol.	https://data.4dnucleome.org/files-processed/4DNFI49A9Z1K/
4DNFIPTEIICR	Hi-C experiments on H1 derived endoderm, differentiated using the 4DN protocol. Cells crosslinked with FA and digested with DpnII.	https://data.4dnucleome.org/files-processed/4DNFIPTEIICR/
4DNFIW5Z7XD3	Hi-C experiments on H1 derived endoderm, differentiated using the protocol from Rene Maehr lab. Cells crosslinked with FA and digested w…	https://data.4dnucleome.org/files-processed/4DNFIW5Z7XD3/

Table 3. List of the 4DN dot calls files used in the examples shown in Figs. 2 and 3, their descriptions and download URLs.

After merging with the list of loops that overlap genes (reciprocal of Query 3), the complete list of human genes and overlapping loops was obtained. Analysis for density estimation was performed in R (RStudio 2022.12.0.353[13,14] and R v4.2.2[14]). Functional annotation of the gene list associated with the chromosomal loop with the highest number of genes per 100kbp loop length used Metascape[15].

Results

HSCLO effectively maps chromosomal locations across five genomic resolution levels: whole chromosome, 1 Mbp, 100 kbp, 10 kbp, and 1 kbp. Over 3.4 million nodes are interconnected to enable detailed mapping and querying within the knowledge graph environment. The hierarchical relationships allow seamless transitions between different scales of genomic data, supporting complex queries such as those involving 1 kbp nodes and their positional relationships within larger chromosomal contexts (Table 1).

HSCLO38 defines chromosomal locations within the GRCh38 release for chr 1–22, X, Y, and mtDNA (Fig. 1). It provides hierarchical relationships across five genomic resolution levels: whole chromosome, 1 megabase pair (Mbp), 100 kilobase pairs (kbp), 10 kbp, and 1 kbp. Each node within these class levels is interconnected to its scale parent and the immediate neighbors on either side to support mapping and association between genomic datasets and features. For example, the 1kbp element HSCLO38:chr1.20517001–20518000 is connected to its "scale parent" at a lower resolution, HSCLO38:chr1.20510001–20520000 through the "`below_1kbp_band`" relationship as well as to its immediate neighbors: its 5′ neighbor HSCLO38:chr1.20516001–20517000 and 3′ neighbor HSCLO38:chr1.20518001–20519000 through "`precedes_1kbp_band`" relationships. Using human genome version GRCh38, the HSCLO38 schema results in 3,431,155 nodes and 6,862,195 relationships (Table 1). Similarly, the summary statistics of GRCh37 are provided in Table 2.

We provide a use case for linking biodata at different resolutions to demonstrate the practical application of HSCLO38 in knowledge organization and discovery. A researcher may be interested in identifying genes found

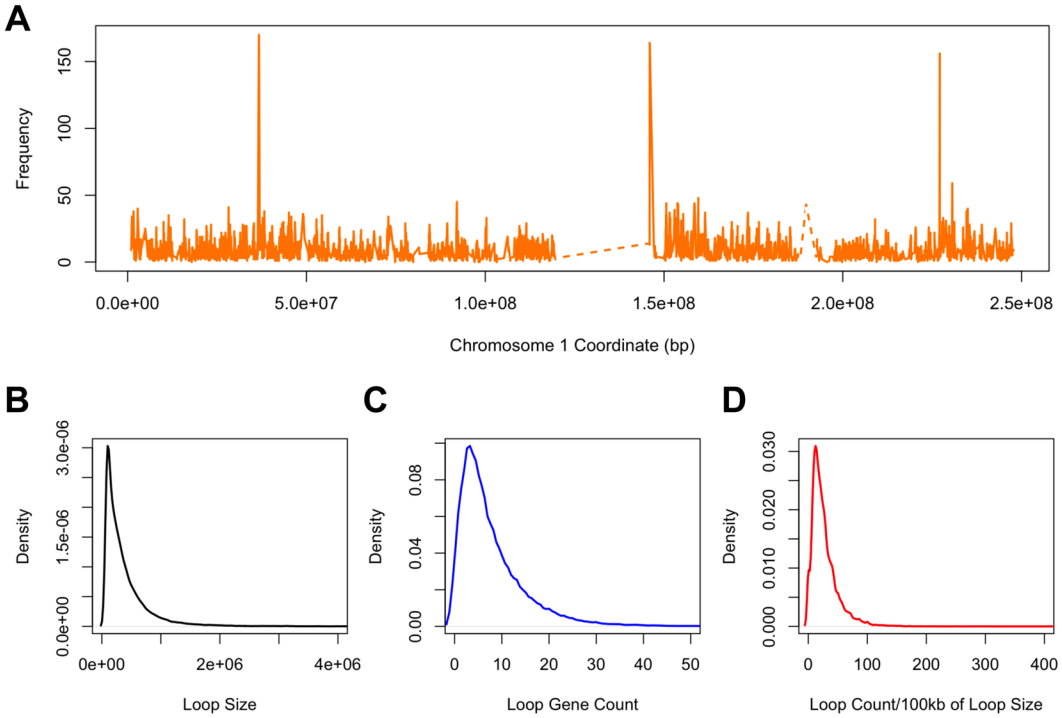

Fig. 3 Use case distributions**.** (**A**) Frequency distribution of genes overlapping the 4DN-computed loops in the 4DN dot call dataset 4DNFIIQP46FO ingested in Petagraph plotted for human chromosome 1 as a function of loop midpoints. (**B**) Loop size distribution according to 4DNFIIQP46FO dataset (in base-pairs), (**C**) gene count per loop distribution and (**D**) distribution of averaged gene count per 100 kbp of loop span derived from the intersection of 4DN loops recorded in 4DNFIIQP46FO and GENCODE genes identified through their connection to HSCLO38.

within large-scale chromatin organization features, such as Hi-C data hosted by the 4DN project[9]. We began by importing HSCLO38 into Petagraph, our custom biomedical KG[7], and then creating edges in the KG to link all gene nodes from GENCODE v41[16] to their respective 1 kbp HSCLO38 nodes. We then created edges for the chromosomal loops from a set of files at the 4DN project[9] to their respective 1 kbp locations in HSCLO38. Using a Cypher query in the Neo4j v5 environment, we retrieved the overlap in 1 kbp nodes between the spans of the GENCODE gene definitions and the start and end points of the 4DN loops. Figure 2 depicts the example query results outlined above. Figure 2A provides a sequence of HSCLO38 connecting the start and end of a 4DN loop upstream anchor. Figure 2B shows how HSCLO38 can extract the corresponding 1 kbp nodes, such as for a human transcript location. Figure 3A illustrates the frequency distribution of human genes overlapping the 4DN loops provided by the 4DN dot call dataset 4DNFIIQP46FO as a function of chromosome 1 coordinates as an example of how HSCLO38 could be utilized to bridge independent datasets in a knowledge graph context. Figure 3 shows the distribution of 4DN loop sizes (Fig. 3B) and the number of GENCODE-defined genes overlapping the 4DN dataset loops in 4DNFIIQP46FO (Fig. 3C,D, respectively). Further analysis of this data reveals ~36,000 loops (from 4DNFIIQP46FO) that overlap at least one gene. To explore the biological relevance of this analysis, we performed functional annotation of the gene list from the loop 4DNFIIQP46FO.chr12.3050000–3060000.chr12.10310000–10320000, which was identified as having the highest number of overlapping genes (173). The analysis provides the top 10 enriched pathways (Table 4), top 10 DisGenNet diseases (Table 5), and top 10 MSigDB cell types (Table 6), implying the disruption in the loop structure and, subsequently, the expression regulation of the overlapping genes could be associated with developmental disorders primarily related to muscular development.

Discussion

The HSCLO is a structured ontological framework essential for organizing and categorizing information on the precise physical positions of genes, genetic markers, and other genomic elements along chromosomes in two human genome versions (GRCh37 and GRCh38). This ontology establishes a standardized vocabulary and hierarchical structure for accurately describing chromosomal positions in knowledge graphs, ensuring uniformity in data representation and sharing across diverse databases and research endeavors. Given the sizable and heterogeneous character of genomic data from multiple sources and studies, HSCLO functions as a unifying framework to enable the integration of datasets based on chromosomal coordinates.

HSCLO offers several advantages that enhance its utility for graphs in biomedical research. The design specifically addresses the challenge of integrating genomic data across multiple resolution levels within knowledge graphs, which supports accurate data alignment across scales, from entire chromosomes to 1kbp segments. This

Pathway ID	Category	Description	Count	%	log10(p)	log10(q)
M5880	Canonical Pathways	NABA ECM AFFILIATED	14	9.86	−13.11	−8.77
GO:0061760	GO Biological Processes	antifungal innate immune response	6	4.23	−9.77	−5.9
GO:0045087	GO Biological Processes	innate immune response	17	11.97	−6.83	−3.28
R-HSA-198933	Reactome Gene Sets	Immunoregulatory interactions between a Lymphoid and a non-Lymphoid cell	7	4.93	−5.49	−2.22
GO:0002831	GO Biological Processes	regulation of response to biotic stimulus	12	8.45	−5.14	−1.91
hsa04610	KEGG Pathway	Complement and coagulation cascades	5	3.52	−4.27	−1.27
WP534	WikiPathways	Glycolysis and gluconeogenesis	4	2.82	−4.23	−1.27
R-HSA-1296072	Reactome Gene Sets	Voltage gated Potassium channels	3	2.11	−2.96	−0.46
WP4172	WikiPathways	PI3K Akt signaling pathway	7	4.93	−2.95	−0.46
GO:0045601	GO Biological Processes	regulation of endothelial cell differentiation	3	2.11	−2.88	−0.43

Table 4. Top 10 pathways associated with genes overlapping loop 4DNFIIQP46FO.chr12.3050000–3060000.chr12.10310000–10320000. Count represents the number of genes overlapping the loop that are associated with the specified ontology term. Percent indicates the percentage of the total genes that are associated with the ontology term (calculations include only genes with at least only one ontology term annotation). Column log10(p) is the p-value expressed in base 10 logarithm, while log10(q) denotes the Metascape provided multi-test corrected p-value.

UMLS ID	DisGenNet Description	Count	%	log10(p)	log10(q)
C0205852	Neoplasms, Embryonal and Mixed	7	4.9	−13	−8.4
C0751365	Cancer, Embryonal and Mixed	7	4.9	−13	−8.4
C0751364	Cancer, Embryonal	7	4.9	−12	−7.8
C0027658	Neoplasms, Germ Cell and Embryonal	8	5.6	−12	−7.6
C0740345	Germ Cell Cancer	8	5.6	−11	−6.7
C0027654	Embryonal Neoplasm	8	5.6	−9.6	−5.8
C4310681	EHLERS-DANLOS SYNDROME, PERIODONTAL TYPE, 2	3	2.1	−7	−3.3
C1290884	Inflammatory disorder	12	8.5	−6.5	−2.9
C2363741	HIV-1 infection	15	11	−6	−2.5
C0008373	Cholesteatoma	7	4.9	−5.4	−2

Table 5. Top 10 DisGenNet abnormalities associated with genes overlapping loop 4DNFIIQP46FO.chr12.3050000–3060000.chr12.10310000–10320000 Count represents the number of genes overlapping the loop that are associated with the specified ontology term. Percent indicates the percentage of the total genes that are associated with the ontology term (calculations include only genes with at least only one ontology term annotation). Column log10(p) is the p-value expressed in base 10 logarithm, while log10(q) denotes the Metascape provided multi-test corrected p-value.

MSigDB Systematic Name	Description	Count	%	log10(p)	log10(q)
M40264	DESCARTES FETAL PANCREAS MYELOID CELLS	8	5.6	−4.8	−1.5
M39193	HAY BONE MARROW CD8 T CELL	5	3.5	−4.7	−1.4
M39050	MANNO MIDBRAIN NEUROTYPES HPERIC	14	9.9	−4.6	−1.4
M40196	DESCARTES FETAL HEART MYELOID CELLS	6	4.2	−4.4	−1.3
M40221	DESCARTES FETAL KIDNEY MYELOID CELLS	7	4.9	−4.2	−1.2
M39116	AIZARANI LIVER C12 NK NKT CELLS 4	4	2.8	−4.1	−1.1
M39266	HU FETAL RETINA MICROGLIA	9	6.3	−4.1	−1.1
M39051	MANNO MIDBRAIN NEUROTYPES HMGL	11	7.7	−4	−1.1
M39306	CUI DEVELOPING HEART C9 B T CELL	6	4.2	−4	−1.1
M41697	TRAVAGLINI LUNG EREG DENDRITIC CELL	11	7.7	−4	−1.1

Table 6. Using HSCLO38, we found 10 single cell types from MSigDB c8 gene sets most associated with the set of genes overlapping 4DN chromosomal loop 4DNFIIQP46FO.chr12.3050000–3060000.chr12.10310000–10320000 Count represents the number of genes overlapping the loo Count represents the number of genes overlapping the loop that are associated with the specified ontology term. Percent indicates the percentage of the total genes that are associated with the ontology term (calculations include only genes with at least only one ontology term annotation). Column log10(p) is the p-value expressed in base 10 logarithm, while log10(q) denotes the Metascape provided multi-test corrected p-value.

capability is particularly valuable when integrating large-scale studies where different datasets must be linked at varying feature resolutions. HSCLO's hierarchical structure also enables rapid querying and efficient data retrieval, which are essential for handling the vast amounts of data typically involved in genomic studies.

Despite these strengths, HSCLO does have certain limitations. The only currently available versions are based on GRCh37 and GRCh38 genome assemblies, which may limit its applicability for other assemblies until an update is made to accommodate additional reference genomes. Another limitation inherent to large reference datasets in graphs is the computational demand associated with maintaining and querying the large number of nodes and relationships, especially at HSCLO's finer resolution of 1 kbp. Large-scale analyses at a finer resolution can be resource-intensive and may pose challenges for researchers working in environments with limited computational infrastructure.

Another potential limitation of employing HSCLO is inherent to the complexity involved in its implementation and use. While HSCLO is designed to facilitate data integration, the initial setup process to annotate a new dataset —such as mapping and linking data coordinates to the ontology—can require data preparation and standardization efforts. To address these challenges, future developments will focus on creating more user-friendly tools and documentation and releasing prepared datasets that simplify the process of using HSCLO.

In clinical and biomedical research contexts, understanding the chromosomal locations of genes associated with diseases or genetic variants assumes critical importance. HSCLO38 and HSCLO37 facilitate the systematic cataloging and classification of such pertinent genetic information, thereby supporting investigations into disease genetics and personalized medicine applications. Furthermore, due to the reliance of computational tools and algorithms on structured data, HSCLO can be a foundational resource for developing robust computational methodologies for genomic analysis and interpretation. Thus, HSCLO can play a pivotal role in harmonizing, integrating, and standardizing genomic data, enhancing data interoperability, fostering interdisciplinary research collaborations, and catalyzing advancements in computational tools essential for fundamental research and applied biomedical applications.

HSCLO stands out from earlier ontologies by focusing on ontologized genomic coordinate binning, facilitating integration across various resolution levels in biomedical knowledge graphs. It addresses the challenge of handling genomic data with differing experimental resolutions while ensuring compatibility with the GRCh38 and GRCh37 genome assemblies. HSCLO can be a valuable tool for researchers and data scientists aiming to integrate and analyze genomic data in large-scale biomedical knowledge environments.

Data availability

Knowledge-graph-ready edgelists (triple format) can be found on the HSCLO project page at the OSF website: https://osf.io/pe8v7/, https://doi.org/10.17605/OSF.IO/PE8V7.

Code availability

The code used to generate and query HSCLO is available in a public repository: https://github.com/TaylorResearchLab/HSCLO/tree/main/HSCLO38.

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References

1. Eilbeck, K. *et al.* The Sequence Ontology: a tool for the unification of genome annotations. *Genome Biol.* **6**, R44 (2005).
2. Baran, J. *et al.* GFVO: the Genomic Feature and Variation Ontology. *PeerJ* **3**, e933 (2015).
3. Mungall, C. J., Emmert, D. B. & FlyBase Consortium. A Chado case study: an ontology-based modular schema for representing genome-associated biological information. *Bioinformatics* **23**, i337–46 (2007).
4. Feng, F. *et al.* GenomicKB: a knowledge graph for the human genome. *Nucleic Acids Res.* **51**, D950–D956 (2023).
5. Church, D. M. *et al.* Modernizing reference genome assemblies. *PLoS Biol* **9**, e1001091, https://doi.org/10.1371/journal.pbio.1001091 (2011).
6. Schneider, V. A. Evaluation of GRCh38 and *de novo* haploid genome assemblies demonstrates the enduring quality of the reference assembly. *Genome Research* **27**, 849–864 (2017).
7. Stear, B. J. *et al.* Petagraph: A large-scale unifying knowledge graph framework for integrating biomolecular and biomedical data. *Sci Data* **11**, 1338, https://doi.org/10.1038/s41597-024-04070-w (2024).
8. NIH Common Fund Data Ecosystem Data Distillery Partnership Repository. *GitHub* https://github.com/nih-cfde/data-distillery.
9. Dekker, J. *et al.* Spatial and temporal organization of the genome: Current state and future aims of the 4D nucleome project. *Mol. Cell* **83**, 2624–2640 (2023).
10. Lonsdale, J. *et al.* The Genotype-Tissue Expression (GTEx) project. *Nat. Genet.* **45**, 580–585 (2013).
11. Heath, A. P. *et al.* Abstract 2464: Gabriella Miller Kids First Data Resource Center: Harmonizing clinical and genomic data to support childhood cancer and structural birth defect research. vol. 79 2464–2464 (American Association for Cancer Research, 2019).
12. James, W. *et al.* The Human Genome Browser at UCSC. *Genome Research* **12**(6), 996–1006, https://doi.org/10.1101/gr.229102 (2002).
13. Posit team. *RStudio: Integrated Development Environment for R.* (Posit Software, PBC, Boston, MA, 2022).
14. R Development Core Team, R. & Others. R: A language and environment for statistical computing (2011).
15. Zhou, Y. *et al.* Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. *Nat. Commun.* **10** (2019).
16. Harrow, J. *et al.* GENCODE: the reference human genome annotation for the ENCODE Project. *Genome Res.* **22** (2012).

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Author contributions

T.M.A. wrote the code, and provided analyses and figures. T.M.A., D.M.T. and B.J.S. wrote and edited the paper. T.M.A., D.M.T. and B.J.S. designed the HSCLO schema. T.M.A. and B.J.S. implemented HSCLO in the knowledge graphs. C.M.N. was the project manager. J.C.S. and J.A.S. were involved in the overall design of the knowledge graph environment around HSCLO and advised on its initial design. D.M.T. conceived of, guided, and funded work on HSCLO.

Competing interests

The authors declare no competing interests.

Additional information

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